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# Key-concept thinking prompting for improved reasoning in large language models

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ABSTRACT

Recent advancements in large language models (LLMs) have significantly accelerated the development of natural language processing (NLP) research, demonstrating remarkable capabilities in understanding and generating human-like text. However, these models face limitations when it comes to system 2 tasks, which require slow, multi-step, and conscious reasoning. To address these limitations, we introduce a method called Key-Concept Thinking (KCT), which enhances the model's reasoning ability by directing it to identify and prioritize key concepts within a problem. Building on the Chain-of-Thought (CoT) prompting method, KCT anchors its approach in core ideas, allowing the model to form a deeper understanding of the problem's structure and purpose. This targeted approach aims to improve both the accuracy and efficiency of the model's reasoning, making it better equipped to handle tasks that require precision and deep understanding. We evaluate our proposed prompting strategies using 24 reasoning tasks across four categories: arithmetic, commonsense, symbolic, and other logical reasoning tasks, with three prominent large models: ChatGLM4, Baichuan2, and DeepSeek, respectively. The results show that the Zero-shot-KCT and Zero-shot-CoT-KCT strategies outperform traditional zero-shot and few-shot prompting strategies, highlighting the effectiveness of incorporating key concept thinking into the reasoning processes of LLMs. Our findings have implications for the development of more effective prompting strategies for LLMs that can handle complex reasoning tasks with higher accuracy and coherence.

## 1. Introduction

Recent advancements in large language models (LLMs) have significantly accelerated the development of natural language processing (NLP) research. These models have demonstrated remarkable capabilities in understanding and generating human-like text across a wide range of tasks. LLMs have enabled breakthroughs in tasks like text generation, machine translation, question answering, summarization, and sentiment analysis, often achieving performance levels that were once thought to be unattainable [1–8].

In LLM inference testing, two types of applications are commonly used: System 1 and System 2. System 1 applications are relatively simple and can be processed intuitively without requiring extensive reasoning. In contrast, System 2 applications are more complex and require multi-step reasoning to reach a solution.

On system 1 tasks, LLMs have shown remarkable abilities. In contrast, when it comes to system 2 tasks, they face limitations. These tasks, often requiring step-by-step logical deduction, sustained attention, or long-term planning, challenge LLMs due to their primarily pattern-based learning approach [9].

To tackle System 2 tasks, two primary approaches have been developed for large language models: fine-tuning and in-context few-shot prompting. Fine-tuning involves training the model on task-specific datasets, adjusting its internal weights to improve performance on a particular domain or problem. This process can be conducted at various levels, from full model fine-tuning—where all parameters are updated, to parameter-efficient methods, which modify only a subset of the model while keeping most weights frozen. Fine-tuning allows the model to internalize task-specific knowledge, leading to more accurate and

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consistent outputs compared to prompting alone. However, it is computationally expensive, often requiring high-end GPUs and extensive hyper-parameter tuning. Moreover, fine-tuning introduces risks such as catastrophic forgetting—where the model loses general knowledge in favor of specialized skills—and overfitting, which can limit its ability to generalize to unseen tasks [2,3].

In-context few-shot prompting leverages a small set of well-chosen examples within the input prompt to guide the model's reasoning process without modifying its internal parameters. This approach exploits the model's ability to recognize patterns from the provided demonstrations and apply them to new but similar problems. Unlike fine-tuning, few-shot prompting does not require retraining, making it highly adaptable across different tasks with minimal setup. However, its effectiveness depends heavily on prompt engineering—selecting informative examples, structuring input effectively, and sometimes using techniques like chain-of-thought prompting to improve multi-step reasoning. Despite these advantages, few-shot prompting has inherent limitations: it lacks memory across interactions, struggles with long-horizon consistency, and may fail to generalize reliably beyond the given examples. Additionally, the model's reliance on implicit pattern matching rather than learned task-specific representations can lead to brittle reasoning, particularly in domains requiring precise logical structure or high factual accuracy [10–12].

To address the limitations of fine-tuning and few-shot prompting, Chain-of-Thought (CoT) prompting method is introduced [10]. This approach provides large language models with examples that demonstrate step-by-step reasoning, guiding the model to adopt a similar incremental approach when tackling reasoning tasks. By structuring prompts in a way that showcases logical, sequential thinking, Chain-of-Thought prompting helps the model break down complex problems into smaller, manageable steps, thereby improving its ability to handle tasks that require detailed reasoning [10,11].

While CoT prompting involves providing LLMs with multi-step reasoning examples to guide them through intricate tasks, Zero-shot-CoT simplifies this process by eliminating the need for example-based prompting. Instead, by adding a simple instruction such as "Let's think step by step" before answering, LLMs can be prompted to use logical, sequential reasoning even without specific examples. Research has shown that this single prompt can dramatically improve LLM performance across various reasoning tasks—including arithmetic, symbolic reasoning, and logical problem-solving—demonstrating that LLMs possess latent zero-shot reasoning capabilities [9]. The development from Chain-of-Thought (CoT) prompting to Zero-shot-CoT represents a significant step in leveraging large language models (LLMs) for complex reasoning tasks.

Filter Supervisor-Self Correction (FS-C) framework [1] is designed to enhance the intermediate reasoning capabilities of Large Language Models (LLMs) when tackling complex problems using Chain of Thought (CoT) prompting. FS-C addresses the challenges of hallucinations and errors in intermediate reasoning by incorporating a multi-perspective approach and a self-correction mechanism.

Chain-of-Thought (CoT) prompting is a technique that emulates the human problem-solving process by encouraging large language models to deconstruct complex reasoning tasks into a series of smaller, manageable steps. This method helps models approach problems incrementally, moving through each part in a logical sequence to reach a solution, much like a person might tackle a challenging multi-step question. CoT has proven effective, but it doesn't capture all aspects of human reasoning. Another crucial dimension of how people think through complex problems is their ability to focus on the most essential concepts within the problem—identifying and honing in on critical points before moving forward. This focused approach often allows people to better understand the structure and requirements of a problem, giving them a clearer path to the solution. Building on this observation, our study introduces a method called Key-Concept Thinking (KCT). KCT enhances the model's reasoning ability by directing it to identify and give priority to key concepts in the problem. By anchoring its approach in these core ideas, the model can form a deeper understanding of the problem's structure and purpose.

With KCT, the model does not just proceed through the reasoning process step by step but does so with greater focus on relevant concepts. This targeted approach is intended to improve both the accuracy and efficiency of the model's reasoning, making it better equipped to handle tasks that require precision and deep understanding. By integrating this concept-driven focus, KCT enhances the model's capacity to approach complex tasks more intelligently and adaptively, leveraging key insights to inform each stage of the reasoning process.

Fig. 1 illustrates six different prompting strategies, each with unique approaches to guiding language models through reasoning tasks. Strategies (a) through (d) are based on established methods from previous research, while strategies (e) and (f) represent novel techniques proposed in this paper. Due to limitations in figure space, only the key concepts identified for strategies (e) and (f) are displayed.

1. Zero-shot prompting (a): This approach involves giving the model a question or task without providing any examples or hints on how to proceed. The model must rely entirely on its pre-existing knowledge, without any contextual guidance, to generate a response.
2. Zero-shot-CoT (b): This is an enhancement of zero-shot prompting that incorporates a "Chain of Thought" (CoT) technique. The model is encouraged to think through each step logically by being prompted to solve the task in a step-by-step manner, even though no specific examples are provided.
3. Few-shot prompting (c): In this strategy, a few examples related to the task are provided to help the model infer the desired pattern or approach. This method leverages the model's ability to learn from context and is particularly effective when minimal data is available.
4. Few-shot-CoT (d): Building on few-shot prompting, this strategy combines example-based guidance with step-by-step reasoning (CoT). It encourages the model to mimic the logical structure demonstrated in the provided examples, making it better suited for tasks requiring complex, multi-step reasoning.
5. Zero-shot-KCT (e) (Ours): Introduced in this paper, Zero-shot-KCT directs the model to identify critical concepts within the task before proceeding. This targeted focus aims to enhance the model's understanding of the task's structure by centering on essential elements that shape the solution.
6. Zero-shot-CoT-KCT (f) (Ours): This method combines Zero-shot-CoT with KCT, encouraging the model to identify key concepts and apply step-by-step reasoning simultaneously. By integrating both logical progression and targeted concept recognition, this approach aims to further improve the model's accuracy and coherence in complex reasoning tasks.

The main contribution of this paper is the following:

1. The paper proposes a novel prompting idea that directs LLMs to identify and prioritize key concepts within a problem before proceeding with reasoning. This idea aims to improve both the accuracy and efficiency of the model's reasoning.
2. The paper introduces two new prompting strategies: Zero-shot-KCT: Directs the model to identify critical concepts before reasoning. Zero-shot-CoT-KCT: Combines step-by-step reasoning (CoT) with key concept identification (KCT).
3. The paper evaluates the proposed strategies on 24 reasoning tasks across four categories: arithmetic, commonsense, symbolic, and other logical reasoning tasks. The results show that Zero-shot-KCT and Zero-shot-CoT-KCT outperform traditional zero-shot and few-shot prompting strategies.

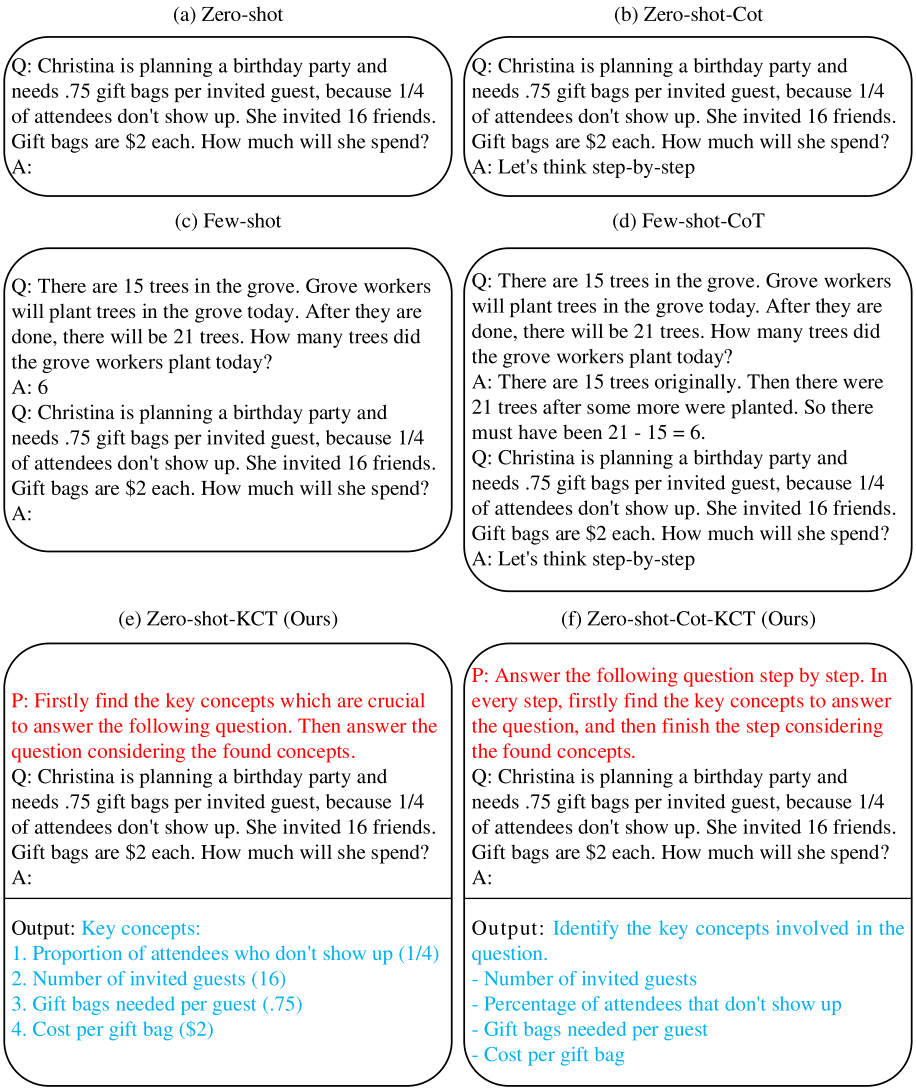


**Fig. 1.** Example inputs and outputs with (a) Zero-shot, (b) Zero-shot-CoT, (c) Few-shot, (d) Few-shot-CoT, (e) Zero-shot-KCT (ours), (f) Zero-shot-CoT-KCT (ours).

## 2. Related work

In recent years, large language models (LLMs) have shown great promise in solving reasoning tasks, such as commonsense reasoning in Chinese [13], arithmetic reasoning [14–19], commonsense reasoning [20,21], symbolic reasoning [10], and other logical reasoning tasks [22]. Two common approaches to leveraging LLMs for these tasks are fine-tuning and few-shot prompting.

Fine-tuning involves training a pre-trained language model on a specific dataset tailored to a particular task. By providing the model with additional task-specific examples and adjusting its parameters, fine-tuning helps the model learn task-relevant patterns and reasoning strategies. This method is particularly effective when large amounts of labeled data are available, as it allows the model to specialize and improve its performance on complex reasoning problems [19,23–25].

On the other hand, few-shot prompting relies on the ability of LLMs to perform reasoning tasks with minimal task-specific data. This technique involves providing the model with a small number of examples (usually just a few) embedded in the prompt. The model then uses these examples to infer the task and apply the learned patterns to solve new problems. Few-shot prompting is particularly beneficial when labeled data is scarce or when rapid adaptation to new tasks is required, as it leverages the model's pre-existing general knowledge and its capacity for contextual learning [1,26–29].

Both techniques have proven effective in addressing reasoning challenges, with fine-tuning excelling in situations where large datasets are available, and few-shot prompting offering flexibility and efficiency in low-data scenarios.

## 3. Key-concept thinking prompting

When a person thinks and responds to a question, they first consider the key concepts involved. By analyzing these fundamental ideas, they gain a clearer understanding of the question's context and nuances. Only after grasping these essential elements do they proceed to form a coherent answer, ensuring that their response is both relevant and accurate. This process allows for a more thoughtful and insightful reply, grounded in an understanding of the core concepts.

Let's take the following question as an example. "Christina is planning a birthday party and needs 0.75 gift bags per invited guest, because 1/4 attendees don't show up. She invited 16 friends. Gift bags are $2 each. How much will she spend?" In answering this question, the concept of "invited guest" is crucial because not everyone who is invited will actually attend the party. Specifically, since 1/4 of the invited guests don't show up, only 3/4 of them will be present. This affects the number of gift bags Christina needs to prepare, as they are based on the expected number of attendees rather than the total invitations sent. Understanding the distinction between invited guests

and actual attendees is essential for calculating the correct quantity of gift bags and, consequently, the total cost.

This approach of focusing on the main ideas highlights the concept of Key-Concept Thinking (KCT) prompting. By identifying and emphasizing critical concepts, such as "invited guests" versus "actual attendees" in this example, Key-Concept Thinking prompting guides us to concentrate on the essential parts of a question before formulating a response. This method not only enhances understanding but also helps ensure that each step in the reasoning process aligns with the underlying concepts, leading to clearer, more accurate answers.

Key-Concept Thinking offers a range of appealing advantages for enhancing reasoning in both human and model responses.

1. KCT helps focus attention on the fundamental ideas in a question, allowing complex problems to be tackled by breaking them down into core components that guide a more relevant and accurate response.
2. KCT is highly adaptable and can be applied across different large language models, regardless of their size or complexity, making it a versatile tool for enhancing reasoning capabilities.
3. KCT is versatile across various domains, including tasks such as logical reasoning, critical reading, and problem-solving, especially where understanding core ideas is crucial to building a solid answer.
4. KCT works seamlessly alongside Chain-of-Thought prompting, where each approach complements the other. While Chain-of-Thought helps to break down a problem into logical steps, KCT ensures that each step remains focused on the core ideas, creating a structured and insightful response. Together, they provide a powerful framework for producing well-reasoned, conceptually sound answers.

To overcome the limitations of few-shot prompting, this paper applies KCT in a zero-shot prompting context, introducing the Zero-shot-KCT strategy. Additionally, recognizing that KCT can be integrated with Chain of Thought (CoT) prompting, the paper also proposes the Zero-shot-CoT-KCT strategy to further enhance zero-shot prompting effectiveness 1.

## 4. Experiment

### 4.1. Configuration

We evaluate our proposed prompting strategy using 24 reasoning tasks across four categories: arithmetic, commonsense, symbolic, and other logical reasoning tasks. Detailed description for every dataset is provided in Section A.2.

In Paper [9], six arithmetic reasoning tasks are employed: SingleEq [14], AddSub [15], MultiArith [16], AQUA-RAT [19], GSM8k [18], and SVAMP [17]. However, since SingleEq and AddSub are relatively simple tasks, large language model (LLM) achieved over 99 % accuracy on them. As a result, we did not include these two tasks in our analysis. Instead, we focused on the four more complex tasks: MultiArith, AQUA-RAT, GSM8k, and SVAMP—which require multi-step reasoning. These tasks provide a more challenging evaluation of the prompting strategies.

For commonsense reasoning tasks, we used two established datasets, CommonsenseQA [20] and StrategyQA [21], along with 14 tasks from the recently introduced CHARM benchmark [13]. CommonsenseQA presents questions with intricate semantics that frequently require drawing on prior knowledge for reasoning. StrategyQA, on the other hand, demands that models infer implicit and multi-hop reasoning to arrive at answers. The CHARM benchmark covers two domains: universally recognized commonsense and China-specific commonsense. It spans seven categories: History (H), Traditional Culture and Arts (CA), Daily Life and Customs (LC), Entertainment (E), Public Figures (F), Geography (G), and the Chinese Language (L).

In symbolic reasoning tasks, we employ Last Letter Concatenation and Coin Flip [10]. In Last Letter Concatenation, the model concatenates the last letters of a set of words, using a random selection of four names per sample. Coin Flip requires the model to determine whether a coin remains heads-up after a sequence of "flip" or "no flip" actions, with samples involving four such trials. While straightforward for humans, language models usually show a flat scaling curve on these tasks.

For additional logical reasoning tasks, we selected two evaluation sets from the BIG-bench project: Date Understanding and Tracking Shuffled Objects [22]. Date Understanding requires models to deduce a date based on provided context. Tracking Shuffled Objects evaluates a model's skill in predicting the final positions of objects after a series of shuffling actions, given their starting positions. In our experiment, we used a dataset involving three objects undergoing shuffling.

Previous research has shown that larger models generally outperform smaller ones in terms of performance. Therefore, this study focuses exclusively on conducting experiments with large-scale models. Specifically, we evaluated three prominent large models: ChatGLM4 [31], Baichuan2 [32], and DeepSeek [33]. The experimental results demonstrate that the proposed prompting strategy is effective across all three models. For simplicity and consistency, the subsequent sections of this paper will report results primarily based on the ChatGLM model.

To evaluate the proposed Zero-shot-KCT and Zero-shot-CoT-KCT prompting strategies, this study compares them against both Zero-shot and Zero-shot-CoT prompts, where Zero-shot serves as the simplest baseline and Zero-shot-CoT represents the highest accuracy among current zero-shot approaches. Additionally, for the MultiArith and GSM8k tasks, the paper extends the evaluation by including comparisons with Few-shot and Few-shot-CoT strategies [10]. In all experiments, we apply greedy decoding consistently across methods. This makes the results for zero-shot approaches deterministic. For few-shot approaches, where the order of in-context examples might influence outcomes [30], we run each experiment once, using a fixed seed across all methods and datasets to ensure fair comparisons with zero-shot methods. Example order did not introduce significant variance when CoT was used in experiments [10].

In this approach, zero-shot prompting strategy is employed, which means that the model's output is not constrained to any specific or predefined format. This inherent flexibility allows the model to generate responses that can vary significantly in structure. For instance, when responding to a multiple-choice question, the model may provide the answer in a variety of ways—such as "[A]", "(A)", or simply "A"—with no guarantee that it will follow the same pattern each time. The diversity in the model's output can be seen not only in the different ways answers can be presented, but also in the language and phrasing used. For example, the model might present an answer as part of a longer sentence or in a completely unstructured form, which further adds to the challenge of answer extraction.

Consequently, it becomes essential to devise a method for extracting the correct answer, regardless of the format in which it appears. When extracting answers from the output of large language models for specific tasks, a structured approach is used to identify the most relevant response. For arithmetic tasks, if the model provides an answer in the form of a range or multiple possibilities, such as "maybe 85 or 86," the first number—"85" in this case—is selected as the final answer. This method prioritizes the first numerical value presented when there is uncertainty or multiple options. For multiple-choice questions, the model might output the answer as a letter, sometimes accompanied by brackets or parentheses, such as "[A]", "(A)", or simply "A." In such cases, the first capital letter is chosen as the answer, ensuring consistency in the selection process. This simple yet effective strategy helps to standardize the extraction of answers across various tasks, minimizing ambiguity and allowing for clear and reliable responses from the model.

### 4.2. Case study

Since the output of LLMs tends to be lengthy and potentially overly engaging, it may inadvertently draw excessive attention from readers. To mitigate this issue and maintain a focused analysis, this section presents a case study limited to three prompting strategies: Zero-shot, Zero-shot-KCT, and Zero-shot-CoT-KCT. The case study is conducted on the following target question:

**Question:** Christina is planning a birthday party and needs 0.75 gift bags per invited guest, because 1/4 of attendees don't show up. She invited 16 friends. Gift bags are $2 each. How much will she spend?

The output of the LLM under the Zero-shot prompt is as follows. Under this prompt, LLM failed to distinguish between "per invited guest" and "per guest," leading to an incorrect answer.

To calculate how many gift bags Christina will need, we first need to determine the number of attendees who are likely to show up, given that 1/4 of the invitees don't show up.

Since she invited 16 friends, 1/4 of them won't show up, which is:

16 friends * 1/4 = 4 friends who won't show up.

So, the number of friends who will likely attend is:

16 friends - 4 friends = 12 friends.

Now, since she needs 0.75 gift bags per guest, we can calculate the total number of gift bags required:

12 friends * 0.75 gift bags per friend = 9 gift bags.

Each gift bag costs $2, so the total cost will be:

9 gift bags * $2 per gift bag = $18.

Therefore, Christina will spend $18 on gift bags.

The output of the LLM under the Zero-shot-KCT prompt is as follows. Under this prompt, LLM successfully identified the key concept "per invited guest" and used it as the basis for its reasoning and calculations. By correctly interpreting this crucial detail, the model was able to produce an accurate answer.

Okay, let's break this down.

Key Concepts:

Number of Invited Guests: This is the starting point for calculations (16 friends).

Gift Bags per Invited Guest: Christina needs a specific number of gift bags for each person invited, regardless of whether they attend (0.75 bags per invited guest).

Cost per Gift Bag: This is the price for each individual gift bag ($2 each).

Attendee No-Show Rate: While the problem mentions that 1/4 don't show up, this information isn't directly used in the calculation for the number of gift bags needed, as the requirement is based on the number of invited guests.

Answer:

Calculate the total number of gift bags needed:

Christina needs 0.75 gift bags for each of the 16 invited guests.

Total gift bags = 0.75 bags/guest × 16 guests = 12 gift bags.

Calculate the total cost:

Each gift bag costs $2.

Total cost = 12 gift bags * $2/gift bag = $24.

Christina will spend $24 on gift bags.

The output of the LLM under the Zero-shot-CoT-KCT prompt is as follows. Under this prompt, LLM successfully identified the key concept "per invited guest" and employed chain-of-thought reasoning to guide its calculations. By breaking down the problem step by step and focusing on the correct interpretation of the term, the model arrived at the correct answer.

Step 1: Identify the key concepts involved in the question.

- Number of invited guests
- Percentage of attendees who don't show up
- Gift bags needed per guest
- Cost per gift bag

Step 2: Calculate the number of attendees who actually show up.

Since 1/4 of attendees don't show up, that means 3/4 do show up. We can represent this as a decimal by dividing 3 by 4, which equals 0.75 (the same as 75 %).

Step 3: Calculate the total number of gift bags needed.

Christina invited 16 friends, and she needs 0.75 gift bags per guest, so we multiply the number of invited guests by the gift bags needed per guest: 16 friends * 0.75 gift bags per friend = 12 gift bags.

Step 4: Calculate the total cost of the gift bags.

Each gift bag costs $2, so we multiply the number of gift bags by the cost per bag:

12 gift bags * $2 per bag = $24

So, Christina will spend $24 on gift bags.

### 4.3. Results

Due to the large number of reasoning tasks used in this paper, a total of 24 tasks, including all the results in a single table would create an excessively long table that could hinder readability. Therefore, the results are presented in two separate tables. One table shows the results for the tasks which are termed as traditional tasks in this paper from Ref. [9]. The other table displays the results for the recently proposed tasks which are termed as CHARM tasks in this paper from Ref. [13].

Table 1 shows the results of traditional tasks, under five prompting strategies, respectively. The highest accuracy for the FS-C prompting strategy is presented under the GPT-4o model. From the table, the following observations can be made:

1. The Zero-shot-KCT strategy shows an improvement in accuracy over the basic Zero-shot strategy in 7 out of 10 tasks. This indicates that directing the model to identify critical concepts before proceeding with reasoning tasks enhances the model's performance.
2. The most notable improvement with Zero-shot-KCT is observed in the 'last letters' task, where accuracy jumps from 9.33 % to 34.01 %. This significant leap suggests that focusing on key concepts is particularly beneficial for tasks that require identification and manipulation of specific elements within a problem.
3. The combination of Zero-shot-CoT and KCT, termed Zero-shot-CoT-KCT, achieves high accuracy across 9 out of 10 tasks, outperforming both the Zero-shot and Zero-shot-KCT strategies. This result underscores the effectiveness of combining step-by-step reasoning with key concept identification.
4. The largest improvement for Zero-shot-CoT-KCT over Zero-shot-CoT is seen in the 'bigbench date' task, where accuracy increases from 66.15 % to 73.42 %. This indicates that the integrated approach is especially effective for complex tasks that require both structured reasoning and a deep understanding of key concepts.

**Table 1**
Accuracy comparison for traditional tasks: 'AR' stands for Arithmetic tasks, 'SR' for Symbolic Reasoning tasks, 'CS' for Common Sense tasks, and 'ORT' for Other Reasoning Tasks.

| Task | | Zero-shot | Zero-shot-KCT | Zero-shot-CoT | FS-C | Zero-shot-CoT-KCT |
|---|---|---|---|---|---|---|
| AR | multiarith | 93.33 | 96.67 | 98.33 | 98.7 | 99.4 |
| | gsm8k | 82.67 | 80.1 | 86.67 | 98.3 | 90.67 |
| | aqua | 61.33 | 74.08 | 66.21 | 75.2 | 73.33 |
| | svamp | 92.0 | 89.0 | 91.0 | 95.5 | 90.0 |
| SR | last_letters | 9.33 | 34.0 | 35.33 | 94.2 | 40.0 |
| | coin_flip | 59.33 | 77.33 | 97.33 | 85.9 | 97.76 |
| CS | commonsensqa | 81.33 | 78.03 | 86.67 | 87.9 | 88.05 |
| | strategyqa | 68.26 | 79.33 | 77.33 | 77.2 | 80.02 |
| ORT | bigbench_date | 45.33 | 68.67 | 66.15 | – | 73.42 |
| | object_tracking | 28.06 | 59.33 | 86.12 | – | 92.67 |

The '-' symbol indicates that the data is not reported for the FS-C strategy.

**Table 2**
Accuracy comparison for CHARM tasks.

| Task | Zero-shot | Zero-shot-KCT | Zero-shot-CoT | Zero-shot-CoT-KCT |
|---|---|---|---|---|
| Chinese-Anachronisms-Judgment | 74.67 | 87.33 | 66.67 | 82.0 |
| Chinese-Movie-and-Music-Recommendation | 68.0 | 72.0 | 74.0 | 80.0 |
| Chinese-Natural-Language-Inference | 81.0 | 79.0 | 82.0 | 77.0 |
| Chinese-Reading-Comprehension | 44.5 | 70.0 | 65.5 | 71.0 |
| Chinese-Sequence-Understanding | 59.0 | 82.0 | 87.0 | 90.0 |
| Chinese-Sport-Understanding | 73.0 | 71.0 | 70.5 | 68.0 |
| Chinese-Time-Understanding | 42.0 | 53.0 | 65.0 | 60.0 |
| Global-Anachronisms-Judgment | 84.0 | 84.0 | 73.33 | 81.33 |
| Global-Movie-and-Music-Recommendation | 70.0 | 70.0 | 58.0 | 76.0 |
| Global-Natural-Language-Inference | 75.0 | 66.0 | 73.0 | 75.0 |
| Global-Reading-Comprehension | 68.0 | 75.0 | 67.5 | 74.0 |
| Global-Sequence-Understanding | 59.0 | 88.0 | 90.0 | 94.0 |
| Global-Sport-Understanding | 75.0 | 77.0 | 72.5 | 72.5 |
| Global-Time-Understanding | 46.0 | 58.0 | 48.0 | 59.0 |

5. Zero-shot-CoT-KCT consistently surpasses Zero-shot-KCT across most tasks, particularly in symbolic and complex reasoning. However, in some arithmetic and common sense tasks, the benefits of CoT reasoning are less pronounced, suggesting that the effectiveness of each approach depends on the nature of the task.
6. Although the FS-C strategy employs a complex framework, Zero-shot-CoT-KCT outperforms it in terms of accuracy on 4 out of 8 tasks.

Table 2 shows the results of charm tasks, under four prompting strategies, respectively. From the table, the following observations can be made:

1. The Zero-shot-KCT strategy shows an improvement in accuracy over the basic Zero-shot strategy in 11 out of 14 tasks. This indicates that directing the model to identify critical concepts before proceeding with reasoning tasks enhances the model's performance.
2. The most notable improvement with Chinese-Reading-Comprehension is observed in the 'last letters' task, where accuracy jumps from 44.21 % to 69.54 %. This significant leap suggests that focusing on key concepts is particularly beneficial for tasks that require identification and manipulation of specific elements within a problem.
3. The Zero-shot-KCT strategy shows an improvement in accuracy over Zero-shot-CoT strategy in 8 out of 14 tasks. It indicates that Zero-shot-KCT can achieve accuracy rates comparable to those of Zero-shot-CoT.
4. The combination of Zero-shot-CoT and KCT, termed Zero-shot-CoT-KCT, achieves high accuracy across 11 out of 14 tasks, outperforming both the Zero-shot and Zero-shot-KCT strategies. This result underscores the effectiveness of combining step-by-step reasoning with key concept identification.
5. The largest improvement for Zero-shot-CoT-KCT over Zero-shot-CoT is seen in the 'Global-Movie-and-Music-Recommendation' task, where accuracy increases from 57.62 % to 75.5 %. This indicates that the integrated approach is especially effective for complex tasks that require both structured reasoning and a deep understanding of key concepts.
6. Zero-shot-CoT-KCT generally surpasses Zero-shot-KCT in tasks requiring structured reasoning, such as sequence understanding and time comprehension. However, Zero-shot-KCT remains competitive in areas like anachronism judgment and reading comprehension, where explicit reasoning might be less critical.

**Table 3**
Accuracy comparison with few-shot prompting strategies.

| Task | Few-shot | Zero-shot-KCT | Few-shot-CoT | Zero-shot-CoT | Zero-shot-CoT-KCT |
|---|---|---|---|---|---|
| multiarith | 92.41 | 96.67 | 99.25 | 98.33 | 99.4 |
| gsm8k | 77.13 | 80.1 | 82.25 | 86.67 | 90.67 |

Upon a thorough analysis of the data presented in both Tables 1 and 2, several key observations emerge that highlight the effectiveness of incorporating key concept thinking (KCT) into the reasoning processes of LLMs:

1. The consistent increase in accuracy across various tasks with the introduction of KCT and CoT-KCT suggests that these methods not only improve accuracy but also contribute to more consistent and coherent reasoning in language models.
2. The comparison between Zero-shot, Zero-shot-KCT, Zero-shot-CoT, and Zero-shot-CoT-KCT provides insights into the incremental benefits of adding structured reasoning (CoT) and key concept focus (KCT) to the prompting strategies.
3. These findings have implications for development of more effective prompting for LLM that can handle complex reasoning tasks with higher accuracy and coherence, potentially leading to improvements in various applications such as question answering, summarization, and decision-making systems.

The following section compares the proposed prompting strategy with existing few-shot approaches, including Few-shot and Few-shot-CoT. Prior research [9] has demonstrated that the performance of few-shot prompting strategies is influenced by both the number and order of examples, with optimal performance achieved using eight examples. Therefore, in our evaluation of few-shot strategies, we consistently use eight examples, maintaining the same examples and order as specified in Paper [9].

Table 3 presents a comparative analysis of the proposed strategies, namely Zero-shot-KCT and Zero-shot-CoT-KCT, against the Few-shot and Few-shot-CoT approaches across two tasks, MultiArith and GSM8K. The findings reveal the following:

1. Both Zero-shot-KCT and Zero-shot-CoT demonstrate superior accuracy compared to the Few-shot strategy in both tasks, indicating the effectiveness of incorporating KCT and CoT into the reasoning process.
2. The integration of CoT with Few-shot strategies results in a mixed performance when compared to Zero-shot-KCT. Specifically, Zero-shot-KCT shows lower accuracy than Few-shot-CoT in both tasks, while Zero-shot-CoT only surpasses Few-shot-CoT in the gsm8k task.
3. The combined application of CoT and KCT in the Zero-shot-CoT-KCT strategy outperforms Few-shot-CoT in both tasks, further validating the synergistic benefits of CoT and KCT in enhancing the performance of reasoning tasks.
4. The results show that while Zero-shot-KCT improves performance compared to standard few-shot prompting, Zero-shot-CoT-KCT consistently achieves superior results. This suggests that the combination of CoT reasoning and KCT is particularly effective for arithmetic problem-solving, outperforming both strategies when applied independently.

### 4.4. Ablation study

Paper [9] explored various categories of prompt strategies, including instructive (encouraging reasoning), misleading (hindering reasoning or promoting reasoning in an incorrect way), and irrelevant (unrelated to reasoning). The study demonstrated that instructive prompts

**Table 4**
Impact of Prompt Length and Structure on Reasoning Outcomes.

| strategy name | prompting strategy |
|---|---|
| long-KCT | Carefully analyze the following question and deeply explore the core concepts, key terms, and potential underlying implications. Based on a thorough understanding of these concepts, consider all relevant factors and provide a comprehensive, accurate, and in-depth answer. |
| short-KCT | Firstly find the key concepts which are crucial to answer the following question. Then answer the question considering the found concepts. |
| long-KCT-CoT | Carefully analyze the following question and deeply explore the core concepts, key terms, and potential underlying implications. Based on a thorough understanding of these concepts, consider all relevant factors and provide a comprehensive, accurate, and in-depth answer. Let's think step by step. |
| short-KCT-CoT | Firstly find the key concepts which are crucial to answer the following question. Then answer the question considering the found concepts. Let's think step by step. |
| Zero-shot-KCT-CoT | Answer the following question step by step. In every step, firstly find the key concepts to answer the question, and then finish the step considering the found concepts. |

**Table 5**
Reasoning Accuracy of Prompting Strategies in GSM8K Task.

| strategy name | accuracy |
|---|---|
| long-KCT | 77.48 |
| short-KCT | 80.1 |
| long-KCT-CoT | 83.44 |
| short-KCT-CoT | 88.08 |
| Zero-shot-KCT-CoT | 90.67 |

effectively enhance reasoning performance. Building on this foundation, our research takes a different perspective by investigating the impact of prompt length on reasoning performance. This approach aims to uncover whether the length of a prompt plays a significant role in guiding reasoning and to provide new insights into optimizing prompt strategies for improved cognitive outcomes.

This study examines several distinct prompting strategies, as shown in Table 4. The long-KCT strategy uses a detailed and elaborate prompt to express KCT, providing a comprehensive framing for the task. In contrast, the short-KCT strategy conveys KCT with a more concise prompt, identical to the Zero-shot-KCT method explored in the previous section.

To investigate combined approaches, the long-KCT-CoT strategy integrates long-KCT with CoT prompting by appending a CoT instruction to the long-KCT prompt. Similarly, short-KCT-CoT pairs the short-KCT prompt with a CoT instruction to encourage step-by-step reasoning. Finally, the Zero-shot-KCT-CoT method represents a carefully crafted integration of short-KCT and CoT strategies, aligning with the approach introduced earlier. These variations aim to uncover how the interplay of prompt length and reasoning scaffolds impacts reasoning outcomes, offering insights into the optimization of prompting techniques.

The results of the five prompting strategies presented in Table 4 for the reasoning task GSM8K are summarized in Table 5. From these results, two key observations can be made:

1. The reasoning accuracy of long-prompt strategies is lower than that of short-prompt strategies.
2. Combining the two strategies through simple concatenation results in lower reasoning accuracy compared to carefully integrating them. This aligns with the first observation, as a simple concatenation effectively creates a longer prompt, leading to reduced reasoning accuracy.

These findings are consistent with common intuition. It is widely understood that large language models (LLMs) have more difficulty understanding long documents compared to shorter ones. A long-prompt strategy essentially requires the LLM to process and understand a lengthy document, while a short-prompt strategy involves processing a much shorter text. Consequently, LLMs tend to exhibit higher accuracy when dealing with shorter prompts, reinforcing the notion that prompt length can significantly impact performance in reasoning tasks.

## 5. Discussion and conclusions

While Key-Concept Thinking (KCT) offers significant advantages in enhancing the reasoning capabilities of large language models (LLMs), there are potential disadvantages to consider in practical applications. Firstly, the reliance on key concepts might lead to overemphasis on certain aspects of a problem, potentially neglecting other important but less obvious factors. This selective focus could result in incomplete or biased reasoning if not carefully managed. Furthermore, the effectiveness of KCT may vary depending on the complexity and nature of the task. For instance, in highly ambiguous or open-ended problems, identifying clear and decisive key concepts might be difficult, limiting the applicability of KCT. Finally, the integration of KCT with other prompting strategies, such as Chain-of-Thought (CoT), requires careful calibration to avoid diminishing returns or even negative impacts on performance.

The dataset used in our current study does not include open-ended or ill-defined problems. In future work, we plan to construct such datasets to further evaluate the generalization abilities of LLMs.

This paper has examined the performance of two prominent prompt strategies in language model applications: Chain-of-Thought (CoT) and Key-Concept Thinking (KCT). Through our analysis, we have shown that both approaches offer distinct advantages in enhancing model performance, particularly in complex reasoning tasks. While CoT excels in promoting step-by-step problem-solving and fostering deeper logical connections, KCT emphasizes the identification and utilization of core concepts, leading to more concise and direct solutions. The comparative evaluation highlights the strengths and limitations of each strategy, suggesting that the optimal approach may depend on the specific task at hand. Future work should explore hybrid models that integrate the strengths of both CoT and KCT, further improving the flexibility and efficacy of language models in diverse domains. Ultimately, understanding when and how to apply these strategies will be key to advancing the field of artificial intelligence and improving its real-world applications.

## CRediT authorship contribution statement

**Minghua Tang:** Writing – original draft, Investigation, Conceptualization. **Chen Bian:** Writing – review & editing, Software, Funding acquisition. **Liming Yang:** Visualization, Validation, Resources, Data curation. **Xueling Zhong:** Project administration, Methodology.

## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGLM in order to polish the writing of the paper. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Details of experimental setup

### A.1. Code

Code is available at https://github.com/sigma218/Zero-shot-KCT

### A.2. Dataset description

Table A.6 provides an overview of the descriptions for all tasks included in this paper.

**Table A.6**
Dataset Overview. The "AF" column indicates the Answer Format, where: N = Number, M = One of Multiple Choices, Y = Yes/No Answer, F = Free Format. The "#S" column represents the number of samples, and the "License" column specifies the dataset's licensing information.

| Task | AF | #S | License |
|---|---|---|---|
| MultiArith | N | 600 | Unspecified |
| GSM8K | N | 1319 | MIT License |
| AQUA-RAT | M | 254 | Apache-2.0 |
| SVAMP | N | 1000 | MIT License |
| CommonsenseQA | M | 1221 | Unspecified |
| StrategyQA | Y | 2290 | Apache-2.0 |
| Date Understanding | M | 369 | Apache-2.0 |
| Shuffled & Objects | M | 750 | Apache-2.0 |
| Last Letters | F | 500 | – |
| Coin Flip | Y | 500 | – |
| Chinese-Anachronisms-Judgment | M | 150 | Apache-2.0 |
| Chinese-Movie-and-Music-Recommendation | M | 50 | Apache-2.0 |
| Chinese-Natural-Language-Inference | M | 100 | Apache-2.0 |
| Chinese-Reading-Comprehension | M | 200 | Apache-2.0 |
| Chinese-Sequence-Understanding | M | 100 | Apache-2.0 |
| Chinese-Sport-Understanding | M | 200 | Apache-2.0 |
| Chinese-Time-Understanding | M | 100 | Apache-2.0 |
| Global-Anachronisms-Judgment | M | 150 | Apache-2.0 |
| Global-Movie-and-Music-Recommendation | M | 50 | Apache-2.0 |
| Global-Natural-Language-Inference | M | 100 | Apache-2.0 |
| Global-Reading-Comprehension | M | 200 | Apache-2.0 |
| Global-Sequence-Understanding | M | 100 | Apache-2.0 |
| Global-Sport-Understanding | M | 200 | Apache-2.0 |
| Global-Time-Understanding | M | 100 | Apache-2.0 |

## Data availability

Data will be made available on request.

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